

Addition and subtraction of polynomials

Remove the brackets, collect the like terms — two steps and nothing more.

LESSON 71–72

2 × 40 MIN

MATEMATIKA 7, §33

S8 2.2

LESSON CLOCK

15'

25'

25'

20'

5'

Adding	15 min
Subtracting	25 min
Longer expressions	25 min
Checking	20 min
Homework	5 min

LEARNING OBJECTIVES

- Add two polynomials.
- Subtract one polynomial from another, changing every sign.
- Simplify an expression containing several polynomials.
- Check an answer by substituting a value.

TEXTBOOK

National	pp. 199–204
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

Adding

A plus before a bracket leaves every sign as it is.

$$(3x^2 - 2x + 5) + (x^2 + 4x - 1) = 4x^2 + 2x + 4$$

STEP	LINE
remove brackets	$3x^2 - 2x + 5 + x^2 + 4x - 1$
x^2	$3x^2 + x^2 = 4x^2$
x	$-2x + 4x = 2x$
constants	$5 - 1 = 4$

WORK ONE DEGREE AT A TIME

All the x^2 terms, then all the x terms, then the constants. Doing them in a fixed order means nothing is missed.

Subtracting

A minus before a bracket changes **every** sign inside it.

$$(3x^2 - 2x + 5) - (x^2 + 4x - 1) = 2x^2 - 6x + 6$$

STEP	LINE
remove brackets	$3x^2 - 2x + 5 - x^2 - 4x + 1$
x^2	$3x^2 - x^2 = 2x^2$
x	$-2x - 4x = -6x$
constants	$5 + 1 = 6$

ALL THREE SIGNS CHANGE, NOT JUST THE FIRST

$-(x^2 + 4x - 1)$ is $-x^2 - 4x + 1$. Writing that line out in full, before collecting, is the single most useful habit in this topic.

Longer expressions

EXPRESSION	ANSWER
$(a + b) + (a - b)$	$2a$
$(a + b) - (a - b)$	$2b$
$(2x - 3) + (x + 1) - (x - 4)$	$2x + 2$
$2(x^2 - x) - 3(x^2 + x)$	$-x^2 - 5x$

The last row combines this lesson with the distributive law: expand first, then collect.

THE DEGREE NEVER RISES

Adding or subtracting polynomials of degree 2 gives a polynomial of degree at most 2 — and sometimes less, if the leading terms cancel.

Checking

Substitute one convenient value into the original and into the answer. If they differ, the simplification is wrong.

CHECK	ORIGINAL AT $x = 1$	ANSWER AT $x = 1$
$(3x^2 - 2x + 5) + (x^2 + 4x - 1)$	$6 + 4 = 10$	$4 + 2 + 4 = 10 \checkmark$
$(3x^2 - 2x + 5) - (x^2 + 4x - 1)$	$6 - 4 = 2$	$2 - 6 + 6 = 2 \checkmark$

USE $x = 1$, OR $x = 2$ IF TERMS CANCEL

At $x = 1$ every power is 1, so the check is a sum of the coefficients — the quickest arithmetic there is. If it happens to give 0 on both sides by accident, try $x = 2$ as well.

02

Worked examples

EXAMPLE 1 Compute $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$.

1

A plus: signs unchanged.

2

$3x^2 - 2x + 5 + x^2 + 4x - 1$

3

Collect by degree.

4

$= 4x^2 + 2x + 4$
Check at $x = 1$: $10 = 10 \checkmark$

ANSWER

$4x^2 + 2x + 4$

EXAMPLE 2 Compute $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$.

1 A minus: every sign in the second bracket changes.

2 $3x^2 - 2x + 5 - x^2 - 4x + 1$

3 Collect by degree.

4 $= 2x^2 - 6x + 6$

Check at $x = 1$: $2 = 2$ ✓

ANSWER

$$2x^2 - 6x + 6$$

EXAMPLE 3 Simplify $2(x^2 - x) - 3(x^2 + x)$.

1 Expand: $2x^2 - 2x$ and $-3x^2 - 3x$.

2 $2x^2 - 2x - 3x^2 - 3x$

3 x^2 : $-x^2$; x : $-5x$.

4 $= -x^2 - 5x$

ANSWER

$$-x^2 - 5x$$

03 Interactive model

Do one subtraction wrongly on purpose — changing only the first sign — and check it at $x = 1$; the check fails, and the class sees why the habit matters.

This model could not start: den is not a function

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sum of polynomials	Ko‘phadlar yig‘indisi	Сумма многочленов
Difference	Ayirma	Разность
To remove brackets	Qavslarni ochish	Раскрыть скобки
Like terms	O‘xshash hadlar	Подобные члены
Opposite polynomial	Qarama-qarshi ko‘phad	Противоположный многочлен
Check by substitution	O‘rniga qo‘yib tekshirish	Проверка подстановкой
Standard form	Standart ko‘rinish	Стандартный вид
Degree	Daraja	Степень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A plus before a bracket:

changes all signs

changes none

changes the first

removes the terms

A minus before a bracket:

changes all signs

changes none

changes the first

doubles the terms

$(a + b) - (a - b)$ equals:

$2a$

$2b$

0

$2a + 2b$

$(3x^2 - 2x + 5) - (x^2 + 4x - 1)$ equals:

The best value for a quick check:

Adding two polynomials of degree 2 gives degree:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(a + b) + (a - b)$

ANSWER $2a$

2. $(a + b) - (a - b)$

ANSWER $2b$

3. $(x + 2) + (x + 3)$

ANSWER $2x + 5$

4. $(x + 2) - (x + 3)$

ANSWER -1

5. $(2x - 1) + (3x + 4)$

ANSWER $5x + 3$

6. $(2x - 1) - (3x + 4)$

ANSWER $-x - 5$

7. A minus before a bracket

ANSWER Changes every sign

Medium · the standard question

7 PROBLEMS

1. $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$

ANSWER $4x^2 + 2x + 4$

2. $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$

ANSWER $2x^2 - 6x + 6$

3. $(2x - 3) + (x + 1) - (x - 4)$

ANSWER $2x + 2$

4. $2(x^2 - x) - 3(x^2 + x)$

ANSWER $-x^2 - 5x$

5. $(a^2 + ab) - (ab - b^2)$

ANSWER $a^2 + b^2$

6. $(5m - 2n) - (3m - 7n)$

ANSWER $2m + 5n$

7. Check $4x^2 + 2x + 4$ at $x = 1$

ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. $(x^3 - 2x^2 + x) - (x^3 + x^2 - 3x)$

ANSWER $-3x^2 + 4x$

2. $3(2a - b) - 2(a - 3b) + (a - b)$

ANSWER $5a + 2b$

3. Subtract $x^2 - 3x + 2$ from $2x^2 + x - 5$

ANSWER $x^2 + 4x - 7$

4. What must be added to $3x - 4$ to give $5x + 1$?

ANSWER $2x + 5$

5. $(a + b + c) - (a - b + c)$

ANSWER $2b$

6. The degree of $(x^2 + x) - (x^2 - x)$

ANSWER 1

7. Simplify $(2x^2 - 3) + (3 - 2x^2)$

ANSWER 0

07 Homework

Homework — 5 tasks

Write the bracket-free line in full, then collect, then check at $x = 1$.

1. Compute $(4x^2 + x - 3) + (2x^2 - 5x + 7)$.
2. Compute $(4x^2 + x - 3) - (2x^2 - 5x + 7)$.
3. Simplify $3(a - 2b) - 2(2a - b)$.
4. Subtract $2y^2 - y + 4$ from $5y^2 + 3y - 1$.
5. What must be added to $7m - 2$ to give $4m + 5$?